Summary in Graph

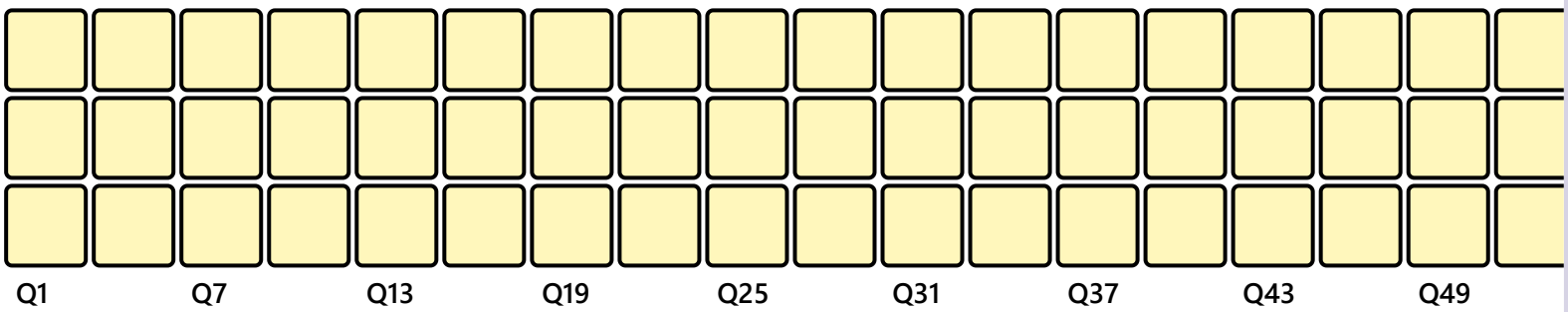
Exam Summary (GO Classes CS 2026 | All India Mock Test 4)

Qs. Attempted:	0 (0 + 0)	Correct Marks:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)	Resultant Marks:	0 (0 + 0)

Total Questions:	65 (30 + 35)
Total Marks:	100 (30 + 70)
Exam Duration:	180 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (71) out of 707 users

EXAM RESPONSE EXAM STATS FEEDBACK

Aptitude

Q #1 Multiple Choice Type Award: 1 Penalty: 0.33 Quantitative Aptitude

A jar contains 600 marbles, each of which is either red, yellow, or green. The ratio of the number of red marbles to the number of yellow marbles to the number of green marbles is $7:3:5$. If 20 marbles of each colour are removed, the new ratio of the number of red marbles to the number of yellow marbles to the number of green marbles is

A. $5:1:3$

B. $15:7:11$

C. $13:5:9$

D. $9:5:7$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Quantitative Aptitude

Alvin, Bingyi, and Cheska play a two-player game that never ends in a tie. In a recent tournament between the three players, a total of 60 games were played and each pair of players played the same number of games.

- When Alvin and Bingyi played, Alvin won 20% of the games.
- When Bingyi and Cheska played, Bingyi won 60% of the games.
- When Cheska and Alvin played, Cheska won 40% of the games.

How many games did Bingyi win?

A. 12

B. 24

C. 30

D. 28

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Quantitative Aptitude

Five numbers $10, 7, 5, 4,$ and 2 are to be arranged in a sequence from left to right following the directions given below:

1. No two odd or even numbers are next to each other.
2. The second number from the left is exactly half of the left-most number.
3. The middle number is exactly twice the right-most number.

Which is the second number from the right?

A. 2

B. 4

C. 7

D. 10

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Quantitative Aptitude

It takes $10\mathrm{s}$ and $15\mathrm{s}$, respectively, for two trains traveling at a different constant speeds to completely pass a telegraph post. The length of the first train is $120\ \mathrm{m}$, and that of the second train is $150\ \mathrm{m}$. The magnitude of the difference in the speeds of the two trains $(\text{in } \mathrm{m/s})$ is

A. 2.0

B. 10.0

C. 12.0

D. 22.0

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Verbal Aptitude

Choose the most appropriate word from the options given below to complete the following sentence:

Under ethical guidelines recently adopted by the Indian Medical Association, human genes are to be manipulated only to correct diseases for which ____ treatments are unsatisfactory.

A. similar

B. most

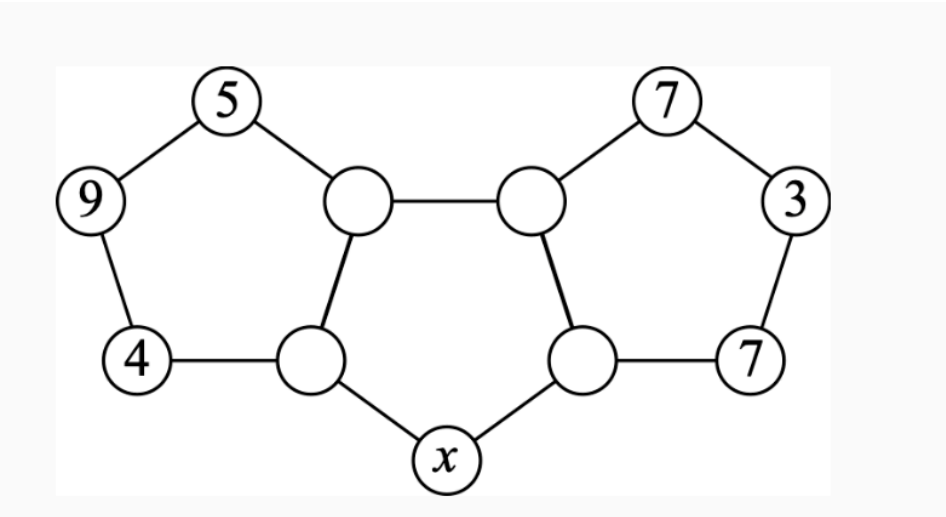
C. uncommon

D. available

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 02sec | Discuss

Q #6 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Quantitative Aptitude

In the diagram shown below, a number is to be written at each of the vertices of the three pentagons. Some of the numbers have already been written.



If the sum of the numbers at the vertices of **each** pentagon is 25, what is the value of x?

A. 7

B. 8

C. 9

D. 10

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Analytical Aptitude

A person purchases two chains from a jeweller, one weighing $18\,\mathrm{g}$ made of 22 carat gold and another weighing $22\,\mathrm{g}$ made of 18 carat gold. Which one of the following statements is correct?

A. 22 carat chain contains $2/11$ times more gold than 18 carat chain

B. 22 carat chain contains $1/11$ times more gold than 18 carat chain

C. Both chains contain the same quantity of gold

D. 18 carat chain contains $2/11$ times more gold than 22 carat chain

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Quantitative Aptitude

A standing (height adjustable) desk has 31 height settings, numbered from the lowest height, 1, to the highest height, 31. Since the desk is not working properly, when the *up* button is pressed, the desk goes up 6 settings at a time if possible; otherwise, it does not move. When the *down* button is pressed, the desk goes down 4 settings at a time if possible; otherwise, it does not move. If the desk starts at setting number 1, how many of the 31 settings will the desk be able to stop at?

A. 14

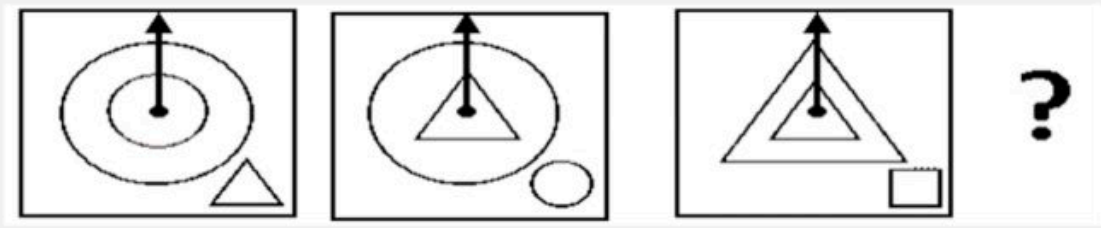
B. 16

C. 9

D. 15

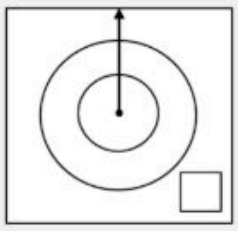
Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Verbal Aptitude

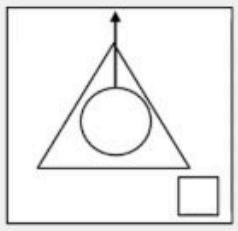


What is the next pattern in the given sequence?

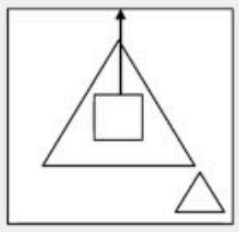
(a)



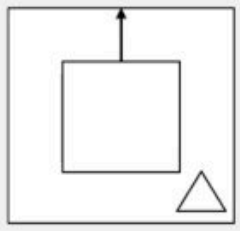
(b)



(c)



(d)



Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Quantitative Aptitude

A chocolate salesman is travelling with 3 boxes with 30 chocolates in each box. During his journey, he encounters 30 tollbooths. Each tollbooth inspector takes one chocolate per box that contains chocolate(s), as tax. What is the largest number of chocolates he can be left with, after passing through all tollbooths?

A. 0

B. 30

C. 25

D. 20

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Technical

Q #1 | Numerical Type | Award: 1 | Penalty: 0 | Computer Networks

To transfer a web page of 9 objects, non-persistent HTTP requires 18 RTTs $(2$ RTT per object$)$.

How many RTTs are needed for persistent HTTP *without pipelining*?

Your Answer: | Correct Answer: 10 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Numerical Type | Award: 1 | Penalty: 0 | Computer Networks

A communication link has a bandwidth of 50 Kbps and a one-way propagation delay of 240 ms. Data is transmitted in fixed-size segments, each of size 1000 bits.

Assume that acknowledgments (ACKs) are of negligible size and that processing delays are negligible.

For a single data segment, let $t = 0$ denote the instant at which the sender begins transmission of that segment. Let T_{RTT} denote the time at which the acknowledgment (ACK) for this segment is received back at the sender.

Consider the open time interval $0 \leq t < T_{\mathrm{RTT}}$. During this interval, the sender may continue transmitting additional data segments.

What is the maximum number of data segments that can be simultaneously in transit, i.e., transmitted by the sender but for which no ACK has yet been received, at any instant t satisfying $0 \leq t < T_{\mathrm{RTT}}$?

Your Answer: | Correct Answer: 25 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Select Type | Award: 1 | Penalty: 0 | Operating System

Page table for process P_1:

Page	Frame
0	0x8d10
1	0x1004
2	0x004a
3	0x5500
4	0x2220
5	0x2221
6	0x2222
7	0x222a
8	0x5558

We consider two virtual addresses: $0\mathrm{x}0001004$ and $0\mathrm{x}0000022$. Two different page sizes may be used:

- 256 bytes (2^8)
- 4096 bytes (2^{12})

Assume that any virtual address whose page number does **not** appear in the table has **no valid translation**. Select **all** correct statements.

A. Under page size 256, the virtual address $0\mathrm{x}0001004$ has **no valid translation**.

B. Under page size 256, the virtual address $0\mathrm{x}0000022$ translates to $0\mathrm{x}8\mathrm{d}1022$.

C. Under page size 4096, the virtual address $0\mathrm{x}0001004$ translates to $0\mathrm{x}1004004$.

D. Under page size 4096, the virtual address $0\mathrm{x}0000022$ translates to $0\mathrm{x}8\mathrm{d}10022$.

Your Answer: | Correct Answer: A;B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Numerical Type | Award: 1 | Penalty: 0 | Operating System

Consider a system in which three jobs *A*, *B*, and *C* arrive for CPU scheduling, starting at time 0.

- Job *A* arrives at time 0 and requires 10 seconds of CPU time.
- Job *B* arrives immediately after job *A* and requires 15 seconds of CPU time.
- Job *C* arrives immediately after job *B* and requires 10 seconds of CPU time.

Assume that the CPU uses a round-robin scheduling policy with the following conditions:

- The time slice (quantum) is unknown, but it is guaranteed to be less than or equal to 2 seconds.
- There is no cost for context switching.
- If one job arrives just before another, it is placed ahead of the later job in the ready queue.

Under these assumptions, at what time does job *B* finish execution?

Your Answer: | Correct Answer: 35 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Programming in C

A farmer must take a lion, a rabbit, and a cabbage across a river.

If the farmer is not present on a side of the river:

- Lion with rabbit → the rabbit is eaten
- Rabbit with cabbage → the cabbage is eaten

In a C program, the variables `lion`, `rabbit`, and `cabbage` are:

- 1 → the item is on the side without the farmer
- 0 → the item is with the farmer

The variable `safe` should be set to 1 if nothing gets eaten, and 0 otherwise.

Which of the following C code is correct?

A.
```
if ((lion && rabbit) || (rabbit && cabbage))
    safe = 0;
else
    safe = 1;
```

B.
```
if ((lion || rabbit) && cabbage)
    safe = 0;
else
    safe = 1;
```

C.
```
if (lion && (rabbit || cabbage))
    safe = 0;
else
    safe = 1;
```

D.
```
safe = !(lion || rabbit || cabbage);
```

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Numerical Type | Award: 1 | Penalty: 0 | Programming in C

Consider the following program segment. The function `pow(x, y)` returns x^y. What is the value of `i` after execution of the program segment?

```
int x = 1, i = 1;
while (x < 1000) {
  x = pow(2, x);
  i = i + 1;
}
```

Your Answer: | Correct Answer: 5 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Programming in C

What does the following function return? Assume that $\mathtt{n} > 0$.

```
int g(int A[], int n)
{
  int i, s;
  for (s = A[0], i = 1; i < n; ++i)
    s = s + A[i] - A[i-1];
  return s;
}
```

A. 0
B. $A[n-1]$
C. $A[n]$
D. $A[n-1] + A[0]$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Algorithms

In Dijkstra's algorithm, define $c(x, y)$ to be the cost between nodes x, y, $D(v)$ the current value of cost of path from source to destination v, $p(v)$ the predecessor node along the path from source to v, and N' the set of nodes whose least cost path is definitely known. Suppose we have $y \in N'$, $D(z) = 10$, $p(z) = y$, and $c(y, z) = 3$. What is the value of $D(y)$?

A. 3

B. 10

C. 7

D. 13

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Algorithms

Let $G = (V, E)$ be an undirected connected simple graph (i.e., no parallel edges or self-loops) with a weight function $w : E \to \mathbb{R}$ on its edge set. Let $w(e_1) < w(e_2) < \cdots < w(e_m)$, where $E = \{e_1, e_2, \ldots, e_m\}$.

Consider the following two statements.

S1. Suppose T is a minimum spanning tree of G with respect to the weight function w. If each edge weight $w(e_i)$ is replaced by $w(e_i)^2$, then T must still be a minimum spanning tree of the modified graph.

S2. Suppose P is a shortest path in G between vertices v and u with respect to the weight function w. If each edge weight $w(e_i)$ is replaced by $w(e_i)^2$, then P is guaranteed to remain a shortest path between v and u.

Which of the following is correct?

A. Both $\mathrm{S1}$ and $\mathrm{S2}$ are true

B. S1 is true and S2 is false

C. S1 is false and S2 is true

D. Both S1 and S2 are false

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Algorithms

Consider the following characters with their frequencies:

Character	Frequency
a	5
b	2
c	8
d	7
e	3
f	7
g	1

A valid Huffman encoding tree is constructed using these frequencies.

Which of the following can be a valid Huffman code for the character e?

A. 0001

B. 01

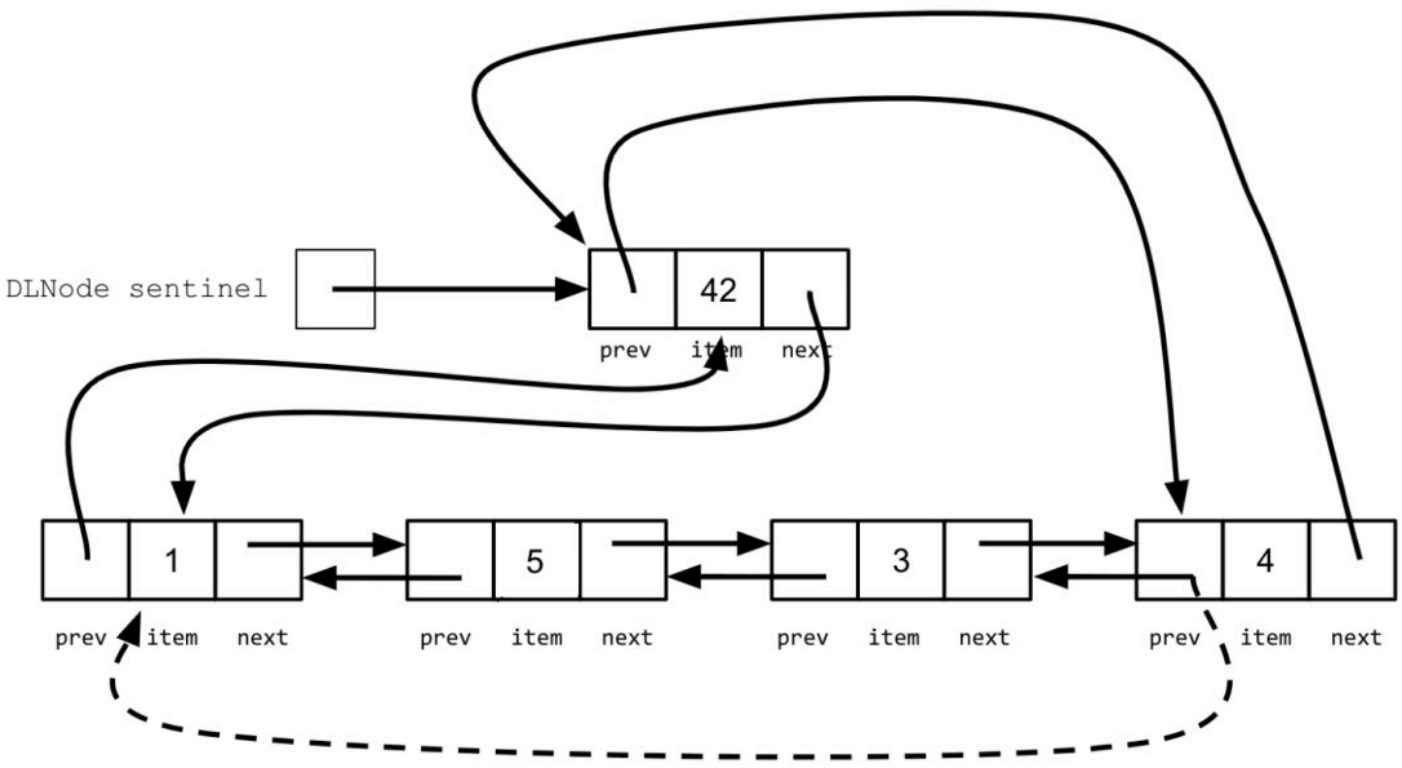
C. 10

D. 11

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Data Structures

Consider a circular doubly linked list as shown in the figure below. The variable `sentinel` is a pointer to the head (sentinel) node of the list. Each node has three fields: `prev`, `item`, and `next`.



Currently, the `prev` pointer of the node containing 4 points to the node containing 3 (shown by the solid arrow). We want to change this `prev` pointer so that it points to the node containing 1 (shown by the dashed arrow).

Which of the following C statements correctly performs this update?

A. `sentinel→prev→prev = sentinel→next;`

B. `sentinel→next→prev = sentinel→prev;`

C. `sentinel→prev = sentinel→next;`

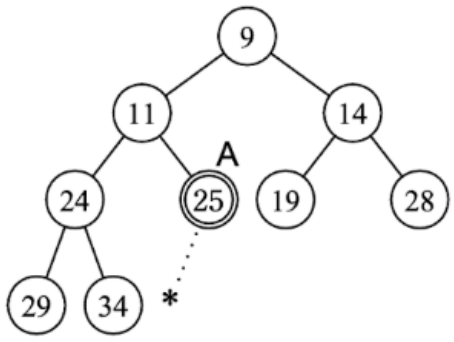
D. `sentinel→prev→next = sentinel→next;`

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12

Multiple Choice Type | Award: 1 | Penalty: 0.33 | Data Structures

In the heap shown below, the value of a parent node is less than the values of child nodes. When inserting a node into this heap, an element is added at the very end. If that element is less than the parent node, the parent and child are exchanged with each other. If element 7 is added to the heap at the position marked by the asterisk $(*)$, what element will end up at position A?



A. 7

B. 11

C. 24

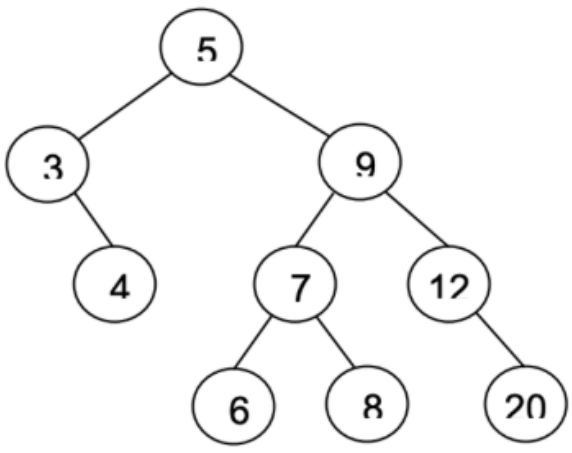
D. 25

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13

Multiple Choice Type | Award: 1 | Penalty: 0.33 | Data Structures

The binary search tree shown below was constructed by inserting a sequence of items into an empty tree.



Which of the following input sequences will not produce this binary search tree?

A. 5 3 4 9 12 7 8 6 20

B. 5 9 3 7 6 8 4 12 20

C. 5 9 7 8 6 12 20 3 4

D. 5 9 3 6 7 8 4 12 20

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

Consider the following matrix:

$$A = \begin{bmatrix} 3 & 0 & 1 \\ 0 & 2 & 0 \\ x & 0 & 1 \end{bmatrix}$$

If all eigenvalues of the matrix are the same, what is the value of x ?

A. 2

B. -2

C. 1

D. -1

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

Let A be an $m \times n$ matrix of rank n with real entries. Choose the correct statement.

A. $Ax = b$ has a solution for any b.

B. $Ax = 0$ does not have a solution.

C. If $Ax = b$ has a solution, then it is unique.

D. $y^T A = 0$ for some nonzero y, where y^T denotes the transpose of the vector y.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #16 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

You have one fair coin and one biased coin which lands Heads with probability $3/4$. One coin is picked uniformly at random and flipped once. What is the probability that the outcome is Heads?

A. $1/2$

B. $3/4$

C. $5/8$

D. $2/3$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #17 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | CO & Architecture

Which instruction set will give the best code density?

A. An accumulator instruction set

B. A RISC instruction set with fixed 32-bit instruction formats

C. A CISC instruction set with variable-length instruction formats

D. A stack instruction set

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #18 | Multiple Select Type | Award: 1 | Penalty: 0 | CO & Architecture

A machine with **register renaming** is able to reorder instructions without regard to what type of dependencies? *(Mark all that apply)*

A. True data dependencies

B. Output dependencies

C. Anti-dependencies

D. Control dependencies

Your Answer: | Correct Answer: B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #19 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Theory of Computation

How many states are in a *smallest DFA* for the language $L_n = (a^n)^*$ if the underlying alphabet is $\Sigma = \{a, b\}$? *(Note: $b^n \notin L_n$.)*

A. n
B. $n+1$
C. $n+2$
D. 1

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #20 Multiple Select Type | Award: 1 | Penalty: 0 | Databases

A database designer observes that a relation R is in 2NF and has transitive functional dependencies. Relation R is then decomposed into relations R_1 and R_2, such that the decomposition is 3NF and lossless. Now, the designer observes that R_2 is in 3NF but not in BCNF. Relation R_2 is further decomposed into R_3 and R_4 such that the decomposition is in BCNF, and it satisfies the following conditions:

$$R_2 = R_3 \cup R_4$$
$$R_3 \cap R_4 \ \rightarrow \ R_3$$

Choose the correct option(s).

A) The decomposition of R_2 into R_3 and R_4 is lossy.

B) The decomposition of R_2 into R_3 and R_4 is lossless.

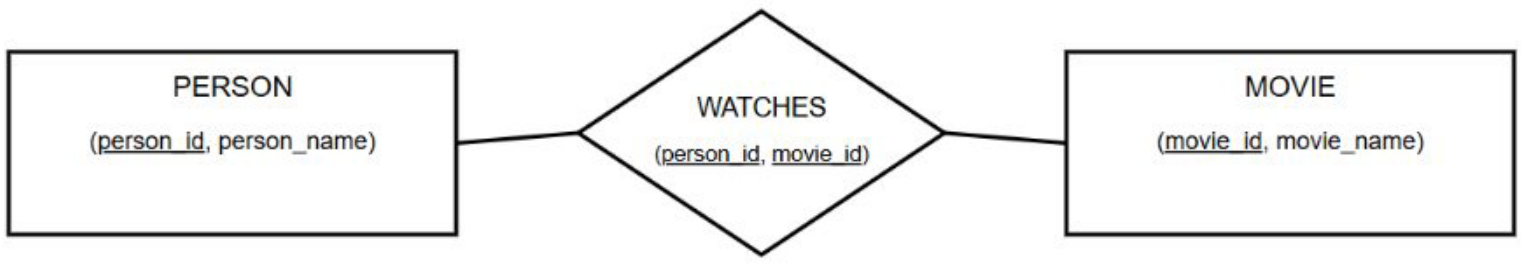
C) $R = R_1 \bowtie R_2$.

D) The number of tuples in R_2 is greater than the number of tuples in $R_3 \bowtie R_4$.

Your Answer: | Correct Answer: B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #21 Multiple Choice Type | Award: 1 | Penalty: 0.33 | Databases

Consider the entity sets *MOVIE* and *PERSON* with a many-to-many relationship set *WATCHES* as shown in the ER diagram below.



If we wish to store the ticket number of a ticket purchased by a person for a movie, then in which among the following must this information appear?

A. PERSON

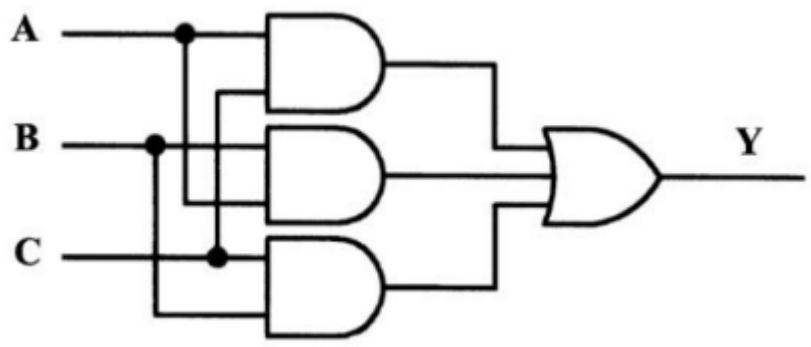
B. WATCHES

C. MOVIE

D. PERSON or MOVIE

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #22 Multiple Choice Type | Award: 1 | Penalty: 0.33 | Digital Logic



The output pulse train Y of the circuit shown on the right, with three synchronized input trains,

$$A = 00001111$$
$$B = 00110011$$
$$C = 01010101$$

will be:

A. 00010111

B. 00100111

C. 01010101

D. 00010001

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #23 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Digital Logic

In a store there are three sensors: a smoke sensor, a motion sensor, and an open-door sensor. Any one of these sensors, when it detects the situation for which it was designed, sends a signal with a high logic level ($A = 1$ for smoke, $B = 1$ for motion, and $C = 1$ for an open door).

However, only the information related to the presence of smoke in the store should automatically activate the fire department, via command Y. Assuming that, in situations where the fire department must be activated, the output corresponding to the command must be at a high logic level ($Y = 1$), select the logical expression that correctly represents the activation of the fire department.

A) $Y = ABC + \overline{ABC}$

B) $Y = A \oplus B \oplus C$

C) $Y = A + B + C$

D) $Y = A$

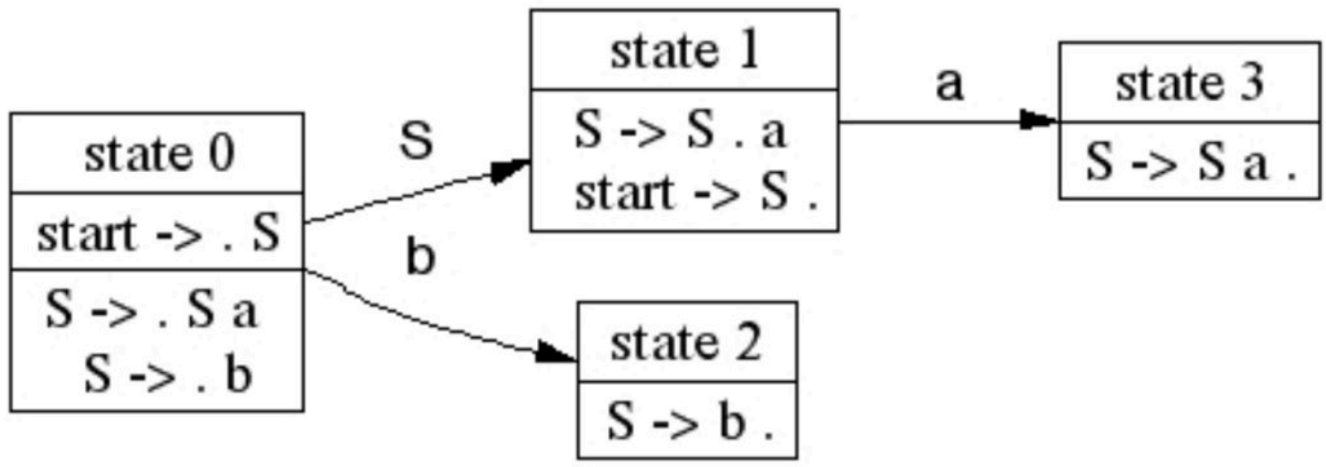
Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #24 | Multiple Select Type | Award: 1 | Penalty: 0 | Compiler Design

Consider the grammar:

```
S → S a | b
```

The NFA of viable prefixes is shown below:



Which of the followings **is/are a viable prefix** of some right-sentential form of this grammar?

A. ba

B. S

C. S a

D. b a a

Your Answer: | Correct Answer: B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #25 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Compiler Design

Consider the following grammar:

$$S \rightarrow X\,a$$
$$X \rightarrow b\,X'$$
$$X' \rightarrow \varepsilon \mid a\,b$$

Assume standard LL(1) parsing with one-token lookahead.

In which situation would the LL(1) parser face a conflict during parsing some string?

A. When expanding X and the next input symbol is b.
B. When expanding X' and the next input symbol is a.
C. When expanding X' and the next input symbol is b.
D. When expanding S and the next input symbol is a.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #26 | Multiple Select Type | Award: 2 | Penalty: 0 | CO & Architecture

Consider three distinct systems, A, B, and C, each equipped with its own cache, denoted A, B, and C respectively.

Cache A is a $16\ \mathrm{Kbyte}$ fully-associative cache.

Cache B is an $8\ \mathrm{Kbyte}$ direct-mapped cache.

Cache C is a $16\ \mathrm{Kbyte}$ two-way set associative cache.

All caches have sixteen-byte lines, start in the same initial state with all lines invalid, use an LRU replacement policy, and see the same sequence of memory references.

After this sequence of memory references, which of the following statements is/are true (mark all that apply)?

A. Cache A will always contain every line in cache B

B. Cache B will always contain every line in cache C

C. Cache C will always contain every line in cache B

D. Cache C will always contain every line in cache A

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #27 | Numerical Type | Award: 2 | Penalty: 0 | CO & Architecture

Suppose you have a CPU with a **6-stage pipeline** containing the following stages:

- **F**: Instruction fetch
- **D**: Instruction decode
- **R**: Register read (and branch address calculation)
- **A**: Execute ALU operations (including determination of branch condition)
- **M**: Memory load and store
- **W**: Write back to register file

Let the **latency** of an instruction be defined as the number of cycles from when this instruction is executed (i.e., completes execution) until the next instruction is executed (i.e., completes execution). For example, the latency of a simple instruction with no dependencies is 1.

Let M be the latency of an **unconditional branch** on this machine (assume no prediction or speculation is employed). Let N be the latency of a **conditional branch** (again assume no prediction or speculation is employed).

What is $M + N$?

(When computing latency of a branch instruction, do not consider the stalls caused by any other instructions except the branch instruction; i.e. assume that the next valid instruction can execute without delay once fetched.)

Your Answer: | Correct Answer: 7 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #28 | Multiple Select Type | Award: 2 | Penalty: 0 | Mathematical Logic

Let $|\alpha|$ denote the size of the formula α. It is defined as:

- $|p| = 0$, where p is an atomic proposition.
- $|\neg\alpha| = 1 + |\alpha|$
- $|\alpha \vee \beta| = 1 + |\alpha| + |\beta|$

Let $\mathrm{sf}(\alpha)$ denote the set of subformulas of the formula α. It is defined as:

- $\mathrm{sf}(p) = \{p\}$, where p is an atomic proposition.
- $\mathrm{sf}(\neg\alpha) = \{(\neg\alpha)\} \cup \mathrm{sf}(\alpha)$
- $\mathrm{sf}((\alpha \vee \beta)) = \{(\alpha \vee \beta)\} \cup \mathrm{sf}(\alpha) \cup \mathrm{sf}(\beta)$

Let $\#(\mathrm{sf}(\alpha))$ denote the number of subformulas of α.

Which of the following is/are true?

A. $\#(\mathrm{sf}(\alpha)) \leq |\alpha| + 1$
B. $\#(\mathrm{sf}(\alpha)) \leq 2|\alpha| + 1$
C. For every $i > 0$, there is at least one formula α_i with $|\alpha_i| = i$ and $\#(\mathrm{sf}(\alpha_i)) = 2|\alpha_i| + 1$
D. For every $i > 0$, there is at least one formula β_i with $|\beta_i| = i$ and $\#(\mathrm{sf}(\beta_i)) < 2|\beta_i| + 1$

Your Answer: | Correct Answer: B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #29 | Multiple Select Type | Award: 2 | Penalty: 0 | Theory of Computation

Which of the following statements is/are true?

A. If L is regular then so is $\{ xx \mid x \in L \}$.
B. If L is regular then so is $\{ xy \mid x, y \in L \}$.
C. If A and B are regular then $\{ xy \mid x \in A \text{ and } y \in B \text{ and } |x| = |y| \}$ is regular.
D. If there's a 5-state NFA that accepts L then there's a 100-state DFA that accepts L.

Your Answer: | Correct Answer: B;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #30 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Theory of Computation

Given a language L, define $\mathrm{perm}(L)$ as the set consisting of all permutations of strings in L. For example, for

$$L = \{abc\}$$

$$\mathrm{perm}(L) = \{abc, acb, bac, bca, cab, cba\}$$

Suppose L is a regular language. Which of the following statements is correct?

A. $\mathrm{perm}(L)$ is always a regular language.

B. $\mathrm{perm}(L)$ is never a regular language.

C. $\mathrm{perm}(L)$ may or may not be regular.

D. $\mathrm{perm}(L)$ is regular if and only if L is finite.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #31 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Databases

Consider that the transactions **T1** and **T2** below may occur simultaneously.

T1	T2
$\mathrm{Read}(A);$	$\mathrm{Read}(B);$
$A = A + 100;$	$\mathrm{Read}(A);$
$\mathrm{Write}(A);$	$\mathrm{Print}(A + B);$
$\mathrm{Read}(B);$	
$B = B - 100;$	
$\mathrm{Write}(B);$	

Analyze the following situations:

I. The operation $\mathrm{Read}(A)$ of **T2** is executed after the operation $\mathrm{Write}(A)$ and before the operation $\mathrm{Read}(B)$ of **T1**. However, the operation $\mathrm{Write}(B)$ of **T1** causes an integrity violation, and transaction **T1** is aborted, with its operations rolled back.

II. After the operations of transaction **T1** have been executed, a message is sent to the user informing that the transaction was successfully completed. However, before the buffers related to **T1** are flushed to the physical storage, a failure occurs, and the data are not effectively recorded.

The transaction properties that were violated in these two situations are, respectively:

A. Atomicity and Consistency.

B. Isolation and Atomicity.

C. Atomicity and Durability.

D. Isolation and Durability.

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #32 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | CO & Architecture

Consider the set of operations described below, written in *register-transfer language*.

$$\begin{aligned}&\text{F1: REM} \leftarrow \text{CP};\\&\text{F2: RDM} \leftarrow M[\text{REM}], \quad \text{CP} \leftarrow \text{CP}+1;\\&\text{F3: RI} \leftarrow \text{RDM};\end{aligned}$$

Data: REM is the memory address register; RDM is the memory data register; RI is the instruction register; CP is the program counter; and $M[X]$ denotes the contents of the memory location identified by address X.

Indicate which phase of instruction processing in a pipelined machine corresponds to the set of operations described above.

A. Instruction decoding

.

B. Operand fetch.

C. Instruction execution.

D. Instruction fetch.

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #33 | Multiple Select Type | Award: 2 | Penalty: 0 | Set Theory & Algebra

In set theory, an urelement or ur-element is an object that is not a set, but that may be an element of a set. It is also referred to as an atom or individual. In this case, if X is an urelement, it makes no sense to say $y \in \mathrm{X}$, although $\mathrm{X} \in \mathrm{U}$ is perfectly legitimate.

Let A be any set in which every element is ur-element.

- We define set B which is union of A and power set of A.

- We define a relation R on B as following : $a\mathrm{R}b$ if and only if $a \in b$.

Which of the following properties is/are satisfied by R?

A. Reflexivity

B. Symmetric

C. Transitive

D. Anti-symmetric

Your Answer: | Correct Answer: C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #34 | Numerical Type | Award: 2 | Penalty: 0 | Databases

Suppose a data file has $1,00,000$ records. Each disk block contains 10 such records (records are unspanned and of fixed length), and the total space required to store the file is $100\ \mathrm{MB}$ (where $\mathrm{M} = 2^{20}$).

A primary index (sparse index with one index entry for every block in the file) is constructed on the data file. The key field is 30 bytes and the pointer field is 10 bytes for each index entry.

What is the minimum number of blocks required to store the index (assuming the index is unspanned)?

Your Answer: | Correct Answer: 39 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #35 | Numerical Type | Award: 2 | Penalty: 0 | Databases

Consider the following database tables, *suppliers* and *parts*.

suppliers

sup_num	sup_name
1001	Able
1002	Peter
1003	Molina
1004	Nikki

parts

part_num	sup_num	part_qty
301	1001	32
301	1004	17
301	1002	41
302	1002	11
302	1003	36
302	1001	16
303	1004	25
304	1002	35
304	1003	40

Let $\{\mathrm{sup_num}\}$ be the primary key of table *suppliers* and $\{\mathrm{part_num}, \mathrm{sup_num}\}$ be the primary key of table *parts*.

Consider the SQL query given below:

```
SELECT s.sup_num, SUM(p.part_qty)
FROM suppliers s, parts p
WHERE p.part_qty > 30
GROUP BY s.sup_num;
```

How many rows will be returned by the above SQL query?

Your Answer: | Correct Answer: 4 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #36 Multiple Select Type | Award: 2 | Penalty: 0 | Mathematical Logic

Suppose $P(x, y)$ is a predicate and the universe for the variables x and y is $\{1, 2, 3\}$. Suppose $P(1, 3), P(2, 1), P(2, 2), P(2, 3), P(3, 1), P(3, 2)$ are true, and $P(x, y)$ is false otherwise.

Which of the following statements is/are false?

A. $\exists x \, \forall y \, P(x, y)$

B. $\neg \exists x \, \exists y \, \big(P(x, y) \wedge \neg P(y, x)\big)$

C. $\forall y \, \exists x \, \big(P(x, y) \rightarrow P(y, x)\big)$

D. $\forall x \, \forall y \, \big((x \neq y) \rightarrow (P(x, y) \vee P(y, x))\big)$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #37 Multiple Select Type | Award: 2 | Penalty: 0 | Graph Theory

Which of the following statements about an undirected graph G with $n > 0$ vertices is/are true?

A. If G is connected, then the number of edges is at least $n - 1$;

B. G is acyclic only if the number of edges is less than n;

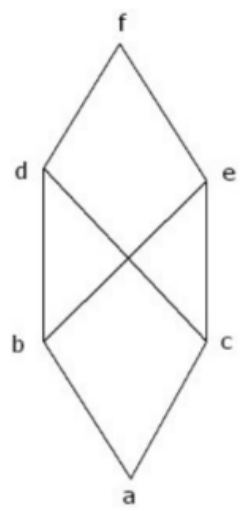
C. If G has no triangles, then G is planar;

D. A connected graph G is Eulerian if and only if every vertex has even degree.

Your Answer: | Correct Answer: A;B;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #38 Multiple Select Type | Award: 2 | Penalty: 0 | Set Theory & Algebra

With respect to the partially ordered set $A(\{a, b, c, d, e, f\}, \leq)$, represented by the Hasse diagram below, Which of the following statements is/are true ?



A. The structure A is not a lattice.

B. The upper bounds of $\{b, c\}$ are the elements d and e.

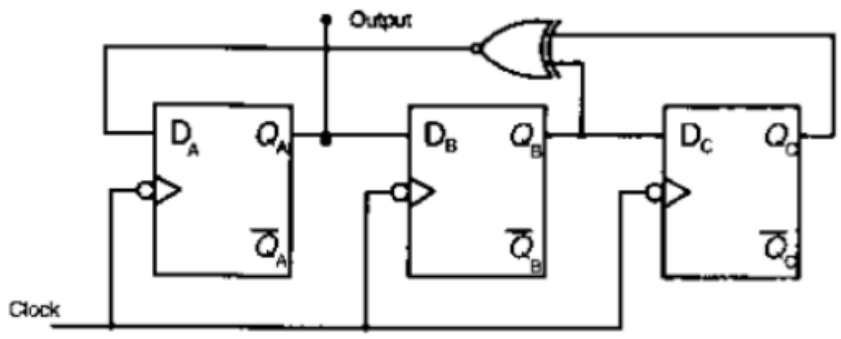
C. The infimum of $\{d, e\}$ is the element a.

D. The structure A has only two totally ordered subsets with 4 elements: $\{a, b, d, f\}$ and $\{a, c, e, f\}$.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #39 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Digital Logic

Assuming that all flip-flops are in reset condition initially, the count sequence observed at $\mathrm{Q_A}$ in the circuit shown is



A. $0010111\ldots$

B. $0001011\ldots$

C. $0101111\ldots$

D. $0110100\ldots$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #40 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Digital Logic

Considering a function described in its canonical sum-of-products form by the minterms $3, 7, 11, 12, 13, 14$, and 15 of a Karnaugh map, and considering variable A as the highest-order logical term, B as the second highest-order term, C as the third highest-order term, and D as the lowest-order logical term, determine its minimized logical representation.

A. $\bar{A}B + CD$

B. $AB + CD$

C. $A\bar{B} + CD$

D. $\bar{A}\bar{B} + CD$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #41 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Compiler Design

Consider the following program that uses the variables `b`, `i`, and `y` :

```
    L1: i = 1
    L2: y = i + 2
    L3: if y > 10 then goto L9
    L4: b = y + 3
 5. L5: if i > 20 goto L7
    L6: i = i + 1
    L7: b = b + i
    L8: goto L2
    L9: y = b + 4
10. L10: return y
```

In the control-flow graph for the program, which variables are live along any of the edges to `L7` ?

A. `b`, `i`, `y`

B. `b`, `i`

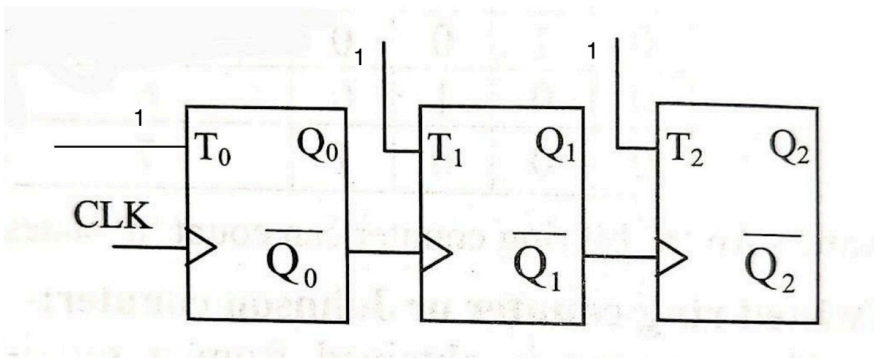
C. `b`, `y`

D. `i`, `y`

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #42 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Digital Logic

The given figure shows a ripple counter using positive edge triggered flip-flops. If the present state of the counter is $Q_2\ Q_1\ Q_0 = 011$, then its next state $(Q_2\ Q_1\ Q_n)$ will be



A. 010

B. 100

C. 111

D. 101

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #43 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Computer Networks

A leaky bucket traffic shaper has a leak rate of 5 packets/sec and a bucket size of 20 packets. The bucket is empty at $t = 0$. Packets arriving when the bucket is full are dropped. After arrivals stop, the bucket continues leaking until it becomes empty.

The packet arrival rates are:

Start (sec)	End (sec)	Arrival rate (pkts/sec)
0	4	6
4	7	10
7	9	8
9	12	4

In the table, the *first row* means that during the time interval from $t = 0$ to $t = 4$ seconds, packets arrive continuously at a rate of 6 packets per second.

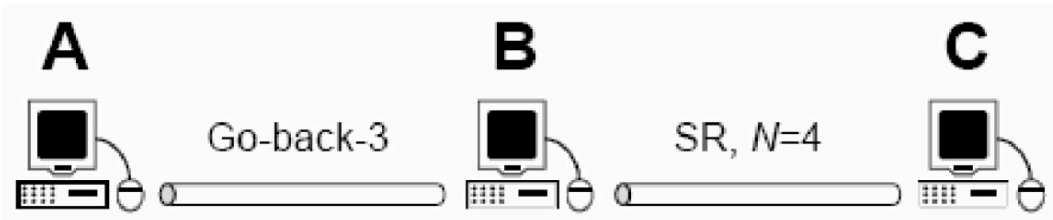
Which option correctly gives the total time (from $t = 0$) for all packets to be transmitted and the total number of packets dropped?

A. 12.4 sec, 20 packets dropped
B. 13.4 sec, 15 packets dropped
C. 14.4 sec, 10 packets dropped
D. 15.4 sec, 5 packets dropped

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #44 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Computer Networks

Sender A transmits packets to intermediate node B using Go-Back-N with a window size of 3. Packets are indexed starting from 0. On the A–B link, packet 1 $(\text{pkt}1)$ and packet 4 $(\text{pkt}4)$ are lost during their first transmission.



Node B forwards packets to receiver C using Selective Repeat with a window size of 4. When B finally receives $\text{pkt}4$ from A and forwards it to C for the first time, $\text{pkt}4$ arrives at C in error, so C does not acknowledge it.

Which packets must B have sent to C *after the first transmission of* $\text{pkt}4$ and *before the timeout caused by* $\text{pkt}4$?

A. None
B. $\{\text{pkt}5\}$ only
C. $\{\text{pkt}5, \text{pkt}6\}$
D. $\{\text{pkt}6\}$ only

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #45 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Computer Networks

A TCP message of size 3000 bytes (including the TCP header) is transmitted from a sender to a destination through two IP routers. The IP header size is 20 bytes, fragmentation offsets are expressed in units of 8 bytes, link-layer headers are ignored for fragmentation, and packets are not reordered.

$$\text{Sender} \xrightarrow{\text{MTU} = 1500} \text{Router 1} \xrightarrow{\text{MTU} = 800} \text{Router 2} \xrightarrow{\text{MTU} = 1000} \text{Destination}$$

Which of the following correctly represents the sequence of IP fragments as they arrive at the destination, in terms of $(\text{total packet length in bytes including IP header, fragment offset})$?

A. $(1500, 0),\ (1500, 185),\ (60, 370)$
B. $(796, 0),\ (724, 97),\ (796, 185),\ (724, 282),\ (60, 370)$
C. $(796, 0),\ (796, 97),\ (724, 194),\ (724, 282),\ (60, 370)$
D. $(796, 0),\ (724, 97),\ (796, 194),\ (724, 291)$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #46 | Numerical Type | Award: 2 | Penalty: 0 | Operating System

Consider a system with three resource types A, B, C and four threads $T1, T2, T3, T4$. The system is currently in the following state.

The total number of resources in the system is $(A, B, C) = (12, 9, 12)$.

Current allocation

Thread	A	B	C
$T1$	2	1	3
$T2$	1	2	3
$T3$	5	4	3
$T4$	2	1	2

Maximum requirement

Thread	A	B	C
$T1$	4	9	4
$T2$	5	3	3
$T3$	6	4	3
$T4$	4	8	2

Available resources

A	B	C
2	1	1

Assume that resources are allocated according to Banker's Algorithm. Any resources granted to thread $T1$ remain allocated until $T1$ finishes execution, and the system must remain in a safe state.

Let (x, y, z) be the maximum number of additional copies of resources (A, B, C) that thread $T1$ can be granted in a single request without risking deadlock.

What is the value of $100x + 10y + z$?

Your Answer: | Correct Answer: 101 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #47 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Operating System

Consider the following modified version of Peterson's algorithm. The variable `turn` is a shared variable accessible to both processes. Before entering the critical region, Process 0 calls `enter(0,1)` and Process 1 calls `enter(1,0)`. Both $\text{interest}[0]$ and $\text{interest}[1]$ are initially 0.

```
void enter(int self, int other) {
    turn = other;              // turn is a shared variable
    interest[self] = 1;
    while (interest[other] == 1 && turn == other);
}

void leave(int self) {
    interest[self] = 0;
}
```

The above differs from the original Peterson's algorithm because turn is assigned *before* $\text{interest}[\text{self}]$.

Which of the following statements is/are correct?

A. The algorithm always ensures mutual exclusion.

B. The algorithm violates mutual exclusion for some interleavings.

C. The algorithm causes deadlock but mutual exclusion holds.

D. The algorithm is starvation-free but not deadlock-free.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #48 | Multiple Select Type | Award: 2 | Penalty: 0 | Operating System

Consider the following concurrent program:

```
int slot = -1;
int array[2] = {0, 0};

function1() {
    P(m);
    slot++;
    V(m);

    int tid = get_counter();

    P(m);
    array[slot] = tid;
    V(m);
}
```

- m is a binary semaphore initialized to 1.
- `slot` and `array` are shared variables.
- Two threads T_1 and T_2 execute `function1()`.
- `get_counter()` returns 1 for T_1 and 2 for T_2.

After both threads finish, which of the following can be the final contents of `array`?

Select all correct options.

A. $[1, 2]$
B. $[2, 1]$
C. $[0, 1]$
D. $[0, 2]$

Your Answer: | Correct Answer: A;B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #49 | Multiple Select Type | Award: 2 | Penalty: 0 | Programming in C

Consider the following C program:

```
typedef struct {
    int x, y;
} Point;

Point* add(Point* a, Point* b) {
    Point res;
    res.x = a->x + b->x;
    res.y = a->y + b->y;
    return &res;
}

int main() {
    Point a = {1, 2};

    Point* b;
    b->x = 3;
    b->y = 5;

    Point* c = malloc(sizeof(Point));
    c = add(&a, b);

    free(b);
    free(c);
}
```

Which of the following statements is/are **correct**?

A. The program has a dangling pointer issue

B. The program contains a memory leak

C. Calling `free(b)` correctly deallocates heap memory

D. Calling `free(c)` correctly deallocates heap memory

Your Answer: | Correct Answer: A;B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #50

Multiple Choice Type | Award: 2 | Penalty: 0.67 | Programming in C

The following recursive function accepts a character array A of size n. Fill in the blank so that the function, when called with $\mathrm{A} = \text{"GATE 2026"}$ and $\mathrm{n} = 9$, changes A to "6202 ETAG".

```
void strFunc (char *A, int n)
{
  char t;
  if (n <= 1) return;
  t = A[0];
  A[0] = A[n-1];
  A[n-1] = t;
  strFunc ( __________);
}
```

A. $\mathrm{A}, \mathrm{n}-1$
B. $\&\mathrm{A}[1], \mathrm{n}-1$
C. $\mathrm{A}+1, \mathrm{n}-2$
D. $\&\mathrm{A}[0], \mathrm{n}-2$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #51

Multiple Select Type | Award: 2 | Penalty: 0 | Algorithms

Consider product of three matrices A, B, and C having size $p \times q$, $q \times r$, and $r \times s$ respectively.

Which of the following statements is/are true?

A. Computing $(AB)C$ and $A(BC)$ both take equal time if $p = q = r = s$.

B. If $(1/q + 1/s) < (1/p + 1/r)$, then $(AB)C$ takes less time than $A(BC)$.

C. $(AB)C$ takes $pr(q + s)$ steps to compute.

D. $(AB)C$ takes less time than $A(BC)$ if $(p + q) > (r + s)$.

Your Answer: | Correct Answer: A;B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #52 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

Suppose you are given a list of n distinct elements such that the position of each element is at most k away from its position in the fully sorted order. The list is sorted using Insertion Sort, and Merge Sort runs in $\Theta(n \log n)$ time. A function $f(n)$ is said to be asymptotically better than a function $g(n)$ if $f(n) = o(g(n))$, where $o(\cdot)$ denotes small-oh.

What is the maximum asymptotic value of k from the given options for which Insertion Sort is asymptotically better than Merge Sort?

A. $k = \Theta(1)$

B. $k = \Theta(\log n)$

C. $k = \Theta\left(\frac{\log n}{\log \log n}\right)$

D. $k = \Theta(n)$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #53 | Numerical Type | Award: 2 | Penalty: 0 | Data Structures

Consider a rooted binary tree with distinct node values, where the tree has a height 10. Note: A tree with only the root has height 0. The tree has the property that its in-order and post-order traversals are same. What is the maximum number of nodes that can be present in such a tree?

Your Answer: | Correct Answer: 11 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #54 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Data Structures

A queue supports the operations $\text{isEmpty}(Q)$, $\text{delete}(Q)$, and $\text{insert}(Q, x)$, where $\text{isEmpty}(Q)$ returns **true** if the queue is empty, $\text{delete}(Q)$ removes and returns the element at the **front**, and $\text{insert}(Q, x)$ adds the element x at the **rear** of the queue.

Consider the following recursive function:

```
void transform(queue Q, int k)
{
  if (k == 0 || isEmpty(Q)) return;
  int t = delete(Q); insert(Q, t);
  transform(Q, k - 1);
}
```

If the queue Q initially contains $[1, 2, 3, 4, 5]$ (front to rear), what will be the contents of Q after the call $\texttt{transform}(Q, 3)$?

A. $[1, 2, 3, 4, 5]$
B. $[4, 5, 1, 2, 3]$
C. $[2, 3, 4, 5, 1]$
D. $[3, 4, 5, 1, 2]$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #55 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus

Suppose that $f(x)$ satisfies Mean Value Theorem on the interval $[1, 3]$. Given that $f(3) = 15$ and $f'(x) \leq 4$ for $1 \leq x \leq 3$. According to the Mean Value Theorem, what is the minimum possible value of $f(1)$?

A. 15

B. 7

C. 3

D. 11

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

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